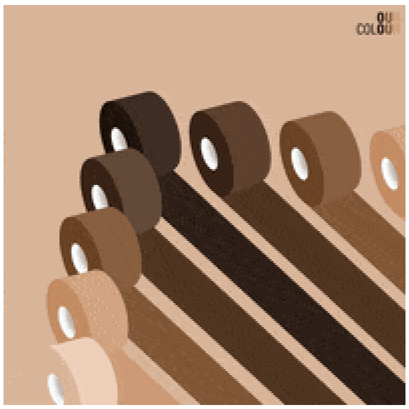
pop quiz rotation inertia

This is a preview of the published version of the quiz

Started: Apr 29 at 7:46pm

Quiz Instructions

based on the lecture



Question 1 15 pts

This person is head spinning at a constant (angular velocity). What will happens when he extends his legs?

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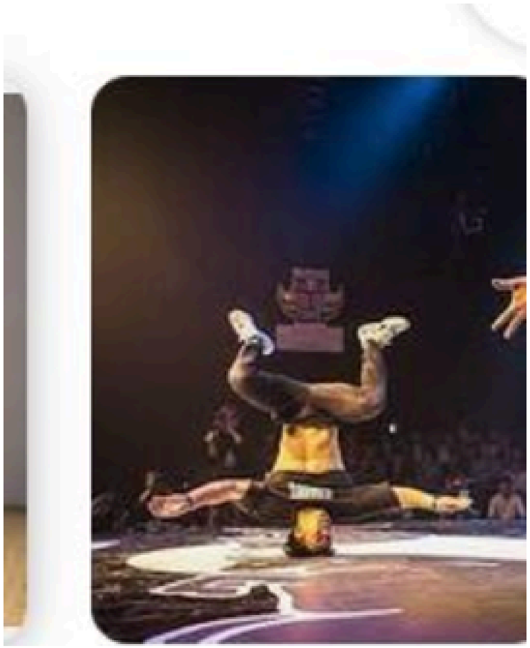


- the angular velocity stays constant
- I missed the class
- He will slow down
- he will speed up

Question 2 12.5 pts

Consider a race between different mass cylinders (solid cylinders) on an inclined plane. The finish line is the end of the slope. We watched a video about this. What is true?

(remember free-fall.)

- The more massive one arrives first.
- the less massive one arrives first.

○
I missed the class

○
They all arrive at the same time.

Question 3 12.5 pts

An airplane has a mass m and is held by a string. It goes around a circle. The radius is R. So the distance between the plane and the axis of rotation is R. It has a moment of inertia I. If you increase the distance by a factor of 3 (so 3R) then the moment of inertia (inertia) increases by a factor of:

as discussed in class. mass stays the same.

○
3

○
2

○
4

○
9

Question 4 15 pts

Ten seconds after an electric fan is turned on, the fan rotates at 300 rev/min. Its average angular acceleration is

(hint: initial angular velocity is 0. convert to rad and to sec)

○
30 rad/s/s

○
30 rev/s/s

○
50 rev/s/s

○
3.14 rad/s/s

Question 5 15 pts

A child pulls on a wagon with a force of 75 N. If the wagon moves a total of 42 m in 3.1 min, what is the average power delivered by the child?

hint: Power = W/t = (Force x distance) / time = Force x (distance/time) = force x speed

- 26W
- 21W
- 10W
- 17W

Question 6 15 pts

A 2.5-kg stone is released from rest and falls toward Earth. After 4.0 s, the magnitude of its momentum is:

hint: take g=9.8m/s/s its an acceleration

- 98kgm/s
- 78kgm/s
- 24kgm/s
- 0kgm/s

Question 7 15 pts

A ball is tied to the end of a cable of negligible mass. The ball is spun in a circle with a radius 2.00 m making 7.00 revolutions every 10.0 seconds. What is the magnitude of the acceleration of the ball?

hint: the speed is constant. there is no tangential acceleration but there is a centripetal acceleration. Because there is a centripetal force to keep the ball on the circular track (provided by the tension). ac=V^2/R

first convert find V(m/s)

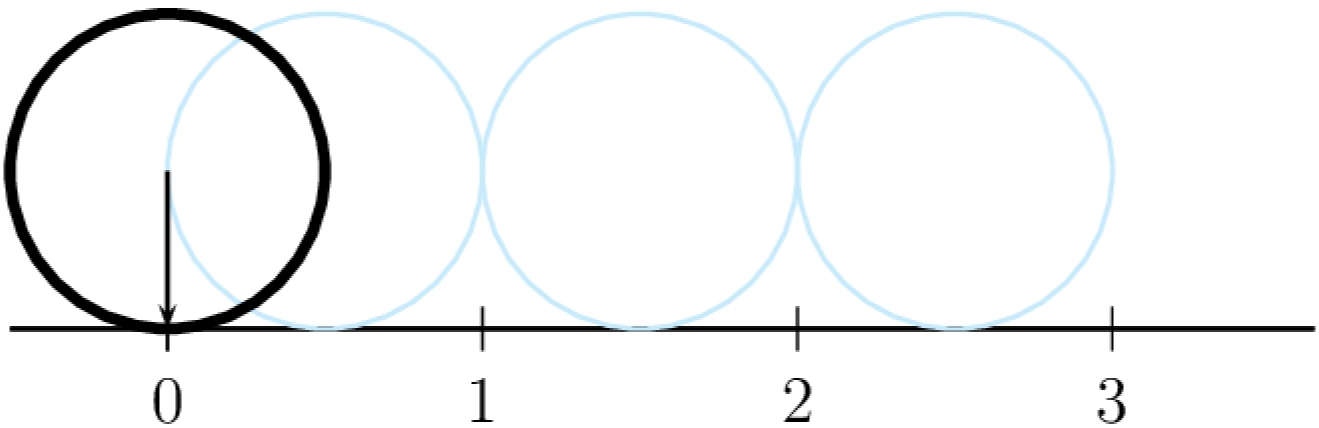
- 38.7m/s/s
- 68m/s/s
- 29m/s/s
- 15m/s/s

○
74m/s/s

Question 8 5 pts

A car tire has a radius of 0.3 meters. If the tire rotates 10 times, how far does the car travel?

think a bit. don't ask me. Imagine the tire is covered with some kind of red ink. For one revolution of the wheel, what is the length of the trace you get on the road ? that is the car has moved by how much ?



○
3m

○
7.5m

○
12.5m

○
19m

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